

Report: Optimising NYC Taxi Operations

Include your visualisations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

1.1.1. Sample the data and combine the files

The following python libraries have been used during the exercise:

- numpy version: 2.5.3
- pandas version: 3.0.6
- matplotlib version: 3.11.2
- seaborn version: 0.13.2
- geopandas version: 1.1.4

Each Yellow Taxi data file (Parquet) contains approximately **3 million records**. To efficiently process this large volume of data, a **5% random sample** was extracted from each file.

Sampling Process:

- Read the monthly data file.
- Iterate through each day in the month.
- For each day, process the data hour by hour.
- Randomly sample 5% of the records for each hour and append them to the sampled dataset.
- Repeat the process for all 24 hours, then move to the next day.
- Continue the same workflow for all days in the month before proceeding to the next monthly file.

After processing all monthly files, the sampled data was consolidated into a single DataFrame representing the entire year, resulting in a dataset with a shape of (3,374,086, 19).

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

The index was reset to create a clean sequential numbering, and the pickup_date and pickup_hour columns were removed as they were no longer required for further analysis.

2.1.2. Combine the two airport_fee columns

- The dataset had two airport fee columns ('airport_fee', 'Airport_fee') with slightly different names.
- These were merged into a single column called airport_fee_combined.
- The first available value from either column was used for each row.
- The original columns were removed to keep the dataset clean and organized.

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

The percentage of missing values was calculated for each column using `df.isnull().mean() * 100`. This helped identify columns with missing data.

```
VendorID          0.000000
tpep_pickup_datetime 0.000000
tpep_dropoff_datetime 0.000000
passenger_count    3.420903
trip_distance      0.000000
RatecodeID        3.420903
store_and_fwd_flag  3.420903
PULocationID      0.000000
DOLocationID      0.000000
payment_type       0.000000
fare_amount        0.000000
extra              0.000000
mta_tax            0.000000
tip_amount         0.000000
tolls_amount       0.000000
improvement_surcharge 0.000000
total_amount       0.000000
congestion_surcharge 3.420903
airport_fee_combined 3.420903
dtype: float64
```

2.2.2. Handling missing values in passenger_count

- Passenger_count is a discrete variable that records the number of passengers per trip.
- Missing values were handled using mode imputation in "Passenger_count" Column.
- The reason to choose the mode is - It provides a meaningful representation of the most typical observation.

2.2.3. Handle missing values in RatecodeID

There are 64874 missing values found in **RatecodeID** and imputed using mode, the first highest frequency (`mode()[0]`).

Since RatecodeID is a categorical variable, using the mode is a practical choice. It retains the most common pattern in the data while avoiding distortion from less frequent categories.

2.2.4. Impute NaN in congestion_surcharge

I filled the missing values in the congestion_surcharge column with the median value. Because this is a numeric feature, the median is a good choice as it represents a typical value and is not influenced by unusually high or low observations.

```
congestion_surcharge
2.5    1690572
0.0    140953
0.5         1
Name: count, dtype: int64
Missing values in 'congestion_surcharge': 64874
Missing values in 'congestion_surcharge after impute':
```

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment type, trip distance and tip amount columns

Outlier Handling Summary:

Payment Type: Records with a payment_type of 0 were removed, as 0 is not a valid payment code. These entries were removed from the dataset.

Trip Distance: Invalid trips with a distance less than 0.1 miles and fares greater than \$300 were removed, as they represent unrealistic fare-to-distance relationships.

Trips with a distance greater than 250 miles and trips with zero distance and zero fare but different pickup and drop-off locations were treated as invalid and removed.

Tip Amount: No records were removed based on tip values. Since tipping is optional, zero tips were retained, while extreme tip values were handled later through feature scaling to reduce their impact on analysis.

3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

3.1.1. **Classify variables** into categorical and numerical

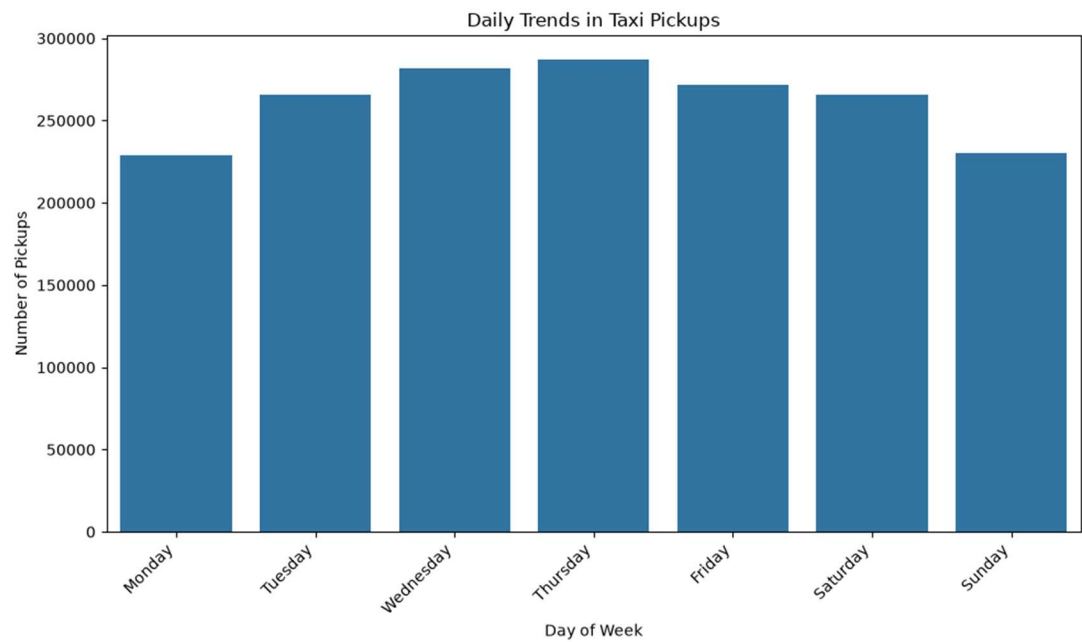
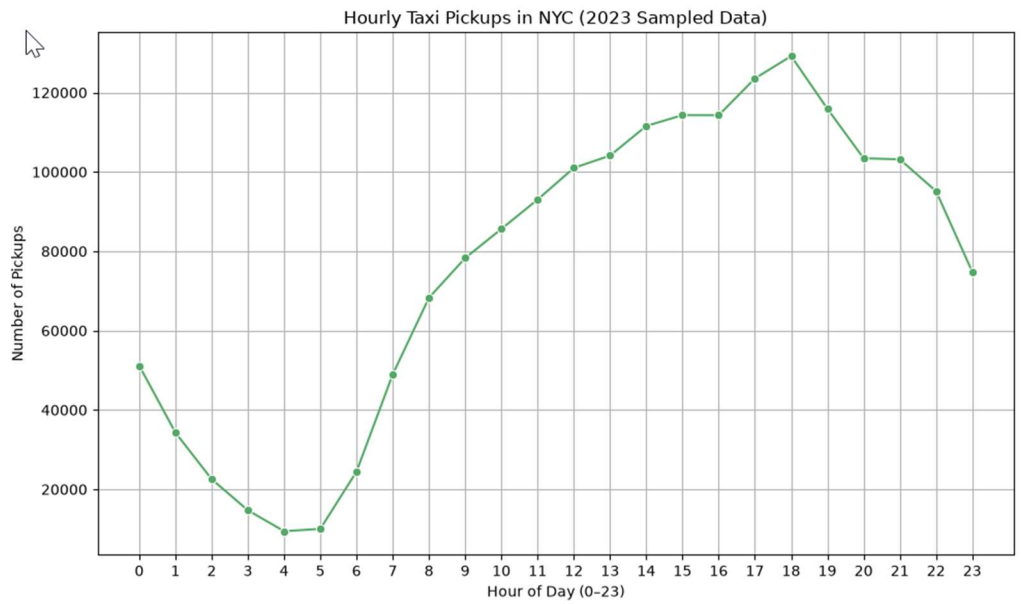
Categorise the variables into Numerical or Categorical.

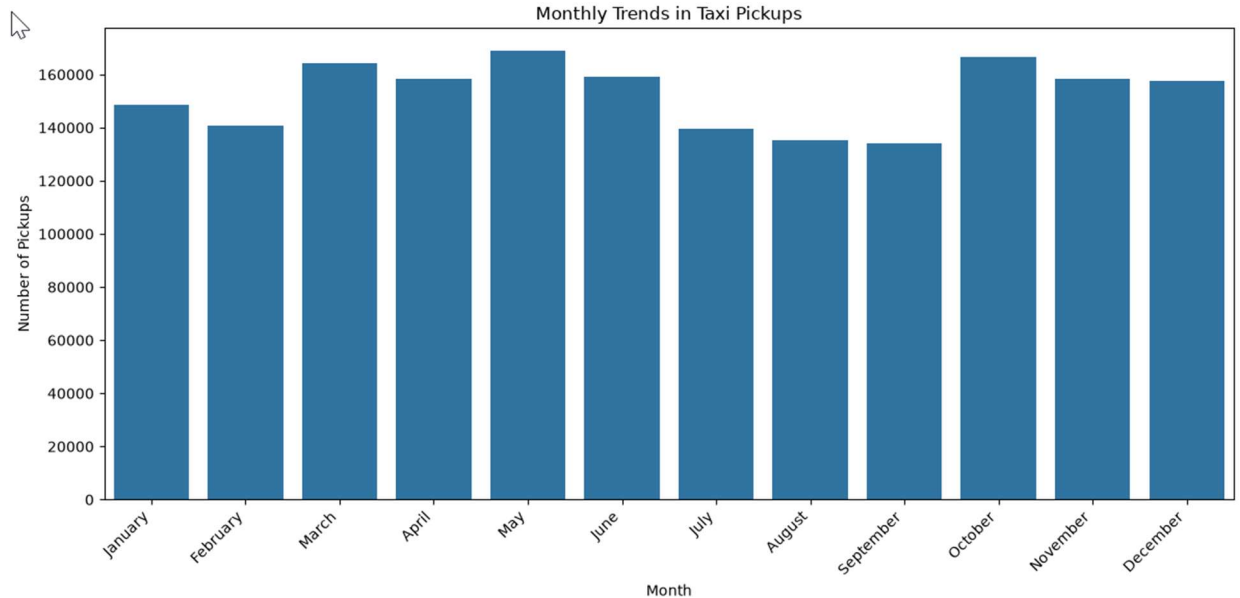
- VendorID:
- tpep_pickup_datetime:
- tpep_dropoff_datetime:
- passenger_count:
- trip_distance:
- RatecodeID:
- PULocationID:
- DOLocationID:
- payment_type:
- pickup_hour:
- trip_duration:

The following monetary parameters belong in the same category, is it categorical or numerical?

- fare_amount
- extra
- mta_tax
- tip_amount
- tolls_amount
- improvement_surcharge
- total_amount
- congestion_surcharge
- airport_fee

3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months





Conclusion

- Taxi demand follows a predictable daily pattern, with the highest pickup volumes occurring during evening rush hours.
- The early morning period (4 AM–5 AM) experiences the lowest demand, while activity increases rapidly after 6 AM.
- Demand remains consistently high throughout the afternoon and evening, reflecting work-related and leisure travel.
- These patterns can be used to improve fleet allocation, reduce passenger wait times, and maximize driver productivity during peak demand periods

3.1.3. Filter out the zero/negative values in fares, distance and tips

To ensure data quality, I filtered out records where:

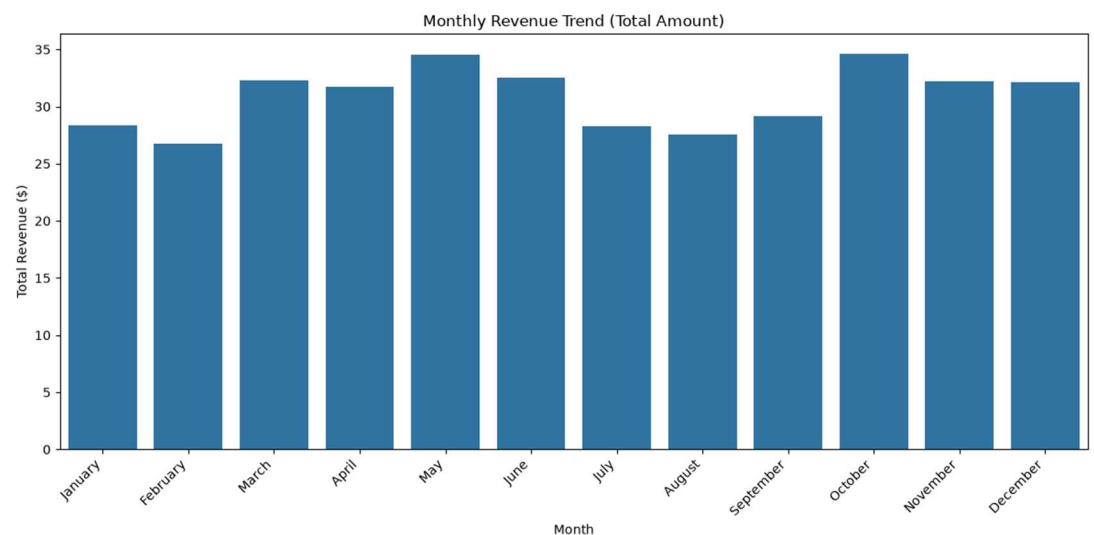
Fare_amount or **total_amount** was zero, as these records are likely to represent invalid, incomplete, or canceled trips.

Trip_distance was zero while the pickup and drop-off locations were different, indicating inconsistent trip information that was removed from the dataset.

Tip_amount values of zero were retained, since tipping is optional and may not include a tip.

This cleaning process improved data quality while preserving realistic passenger behaviour, such as choosing not to tip.

3.1.4. Analyse the monthly revenue trends

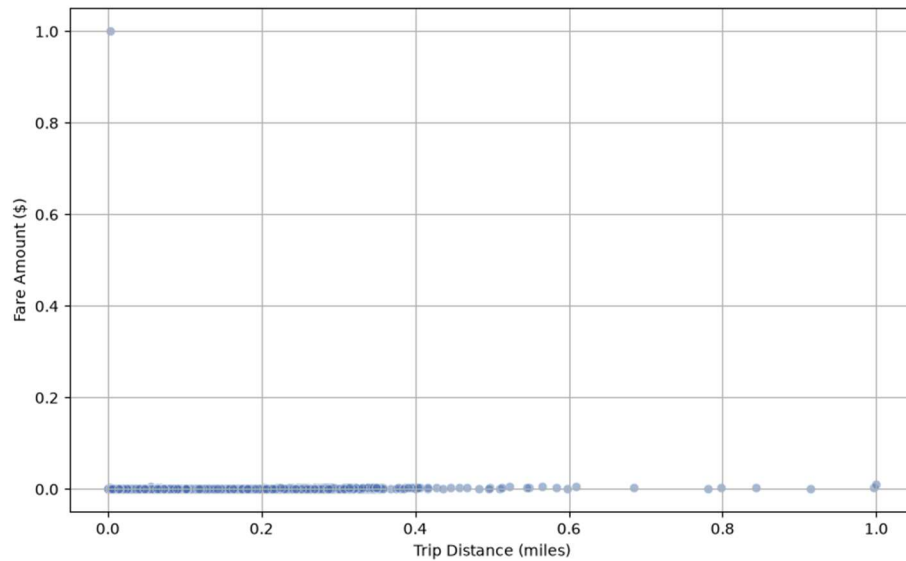


- Revenue remained relatively stable throughout the year, with May and October recording the highest revenue, while February and August showed the lowest revenue levels.
- The trend suggests stronger taxi demand during the spring and fall months, whereas mid-summer months experienced a noticeable dip in overall revenue.

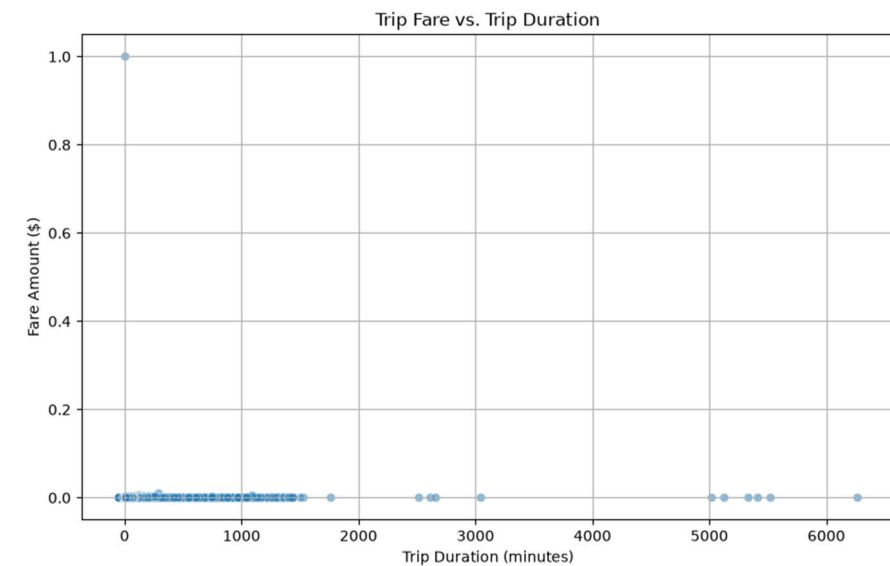
3.1.5. Find the proportion of each quarter's revenue in the yearly revenue

```
pickup_quarter
2022Q4      0.00
2023Q1     23.61
2023Q2     26.68
2023Q3     22.97
2023Q4     26.74
Freq: Q-DEC, Name: total_amount, dtype: float64
```

3.1.6. Analyse and visualise the relationship between distance and fare amount



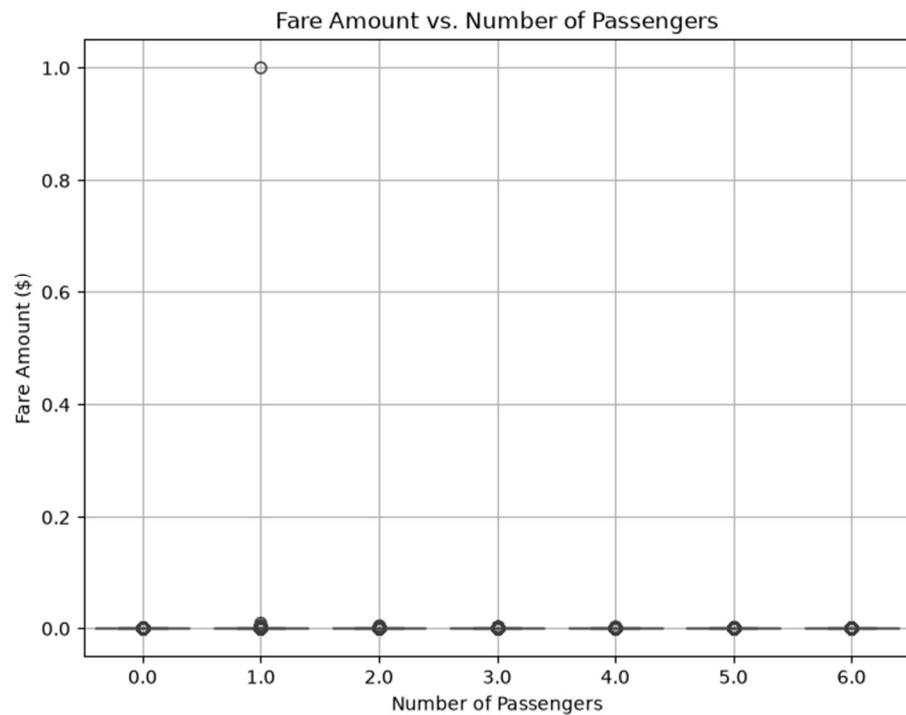
3.1.7. Analyse the relationship between fare/tips and trips/passengers



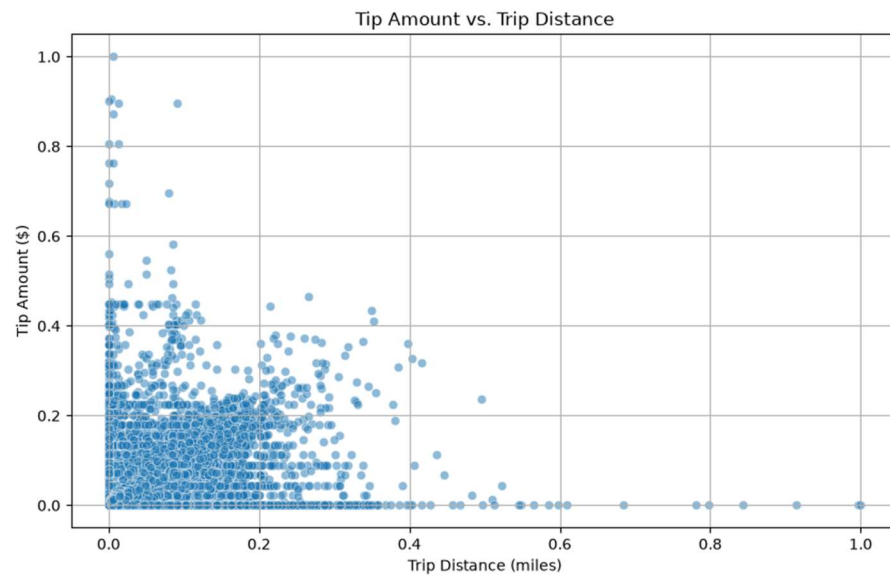
Correlation between trip duration and fare amount: 0.04

- The scatter plot shows a positive relationship between trip distance and fare amount, indicating that fares generally increase as trip distance increases.

- Most trips are concentrated at short distances and lower fare amounts, while only a few long-distance trips generate higher fares, making them relatively less common.



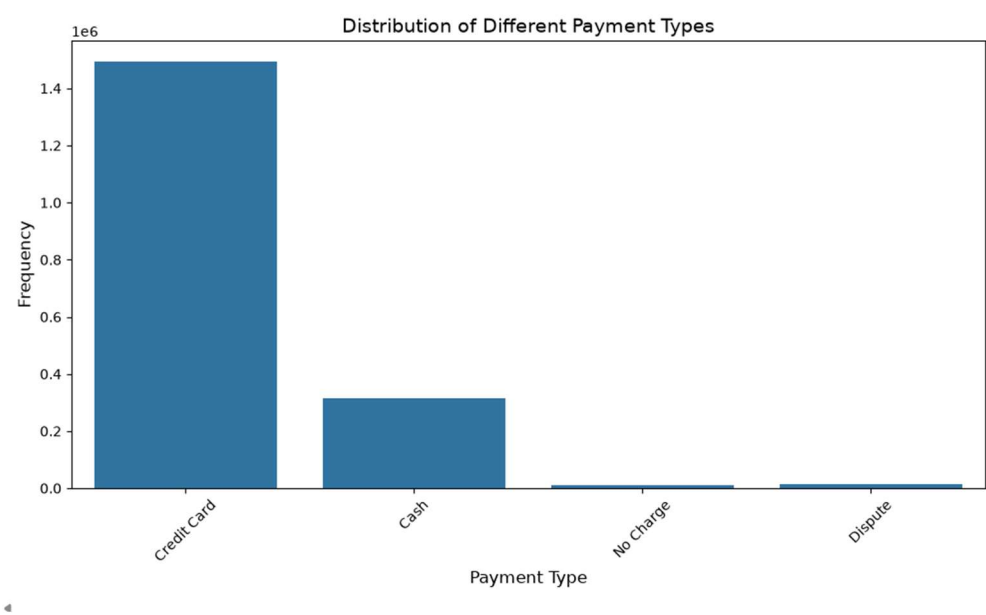
Correlation between passenger count and fare amount: 0.01



Correlation between trip distance and tip amount: 0.57

The plot indicates a weak positive relationship between trip distance and tip amount,

3.1.8. Analyse the distribution of different payment types

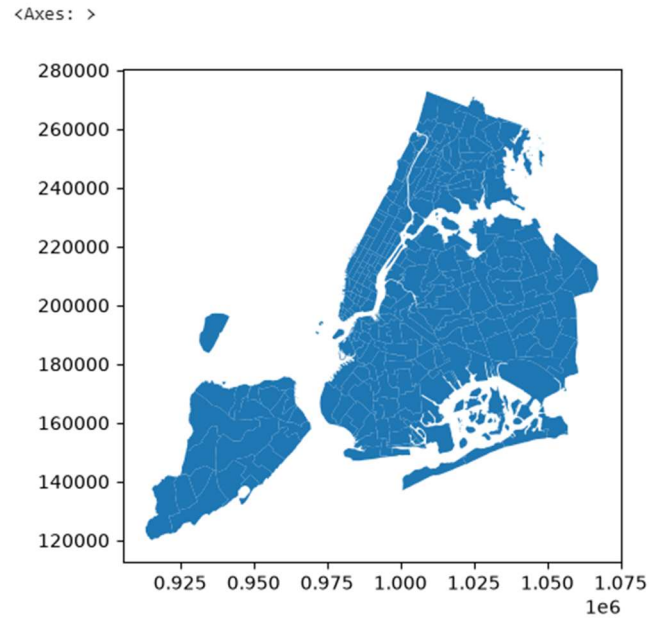


Observation:

- Credit card payments dominate the dataset, accounting for the vast majority of trips, indicating a strong preference for digital payments among NYC taxi passengers.
- Cash remains the second most common payment method, while transactions categorized as No Charge or Dispute represent only a very small fraction of total trips.

3.1.9. Load the taxi zones shapefile and display it

OBJECTID	Shape_Leng	Shape_Area	zone	LocationID	borough	geometry	
0	1	0.116357	0.000782	Newark Airport	1	EWB	POLYGON ((933100.918 192536.086, 933091.011 19...
1	2	0.433470	0.004866	Jamaica Bay	2	Queens	MULTIPOLYGON (((1033269.244 172126.008, 103343...
2	3	0.084341	0.000314	Allerton/Pelham Gardens	3	Bronx	POLYGON ((1026308.77 256767.698, 1026495.593 2...
3	4	0.043567	0.000112	Alphabet City	4	Manhattan	POLYGON ((992073.467 203714.076, 992068.667 20...
4	5	0.092146	0.000498	Arden Heights	5	Staten Island	POLYGON ((935843.31 144283.336, 936046.565 144...



3.1.10. Merge the zone data with trips data

The zones dataset was merged with the trip dataset using locationID and PULocationID as the matching keys.

VendorID	tpep_pickup_datetime	tpep_dropoff_datetime	passenger_count	trip_distance	RatecodeID	store_and_fwd_flag	PULocationID	DOLocationID	payment_type	...	pickup_mor	
0	2	2023-01-01 00:07:18	2023-01-01 00:23:15	1.0	0.037782	1.0	N	138	256	2	...	Janu.
1	2	2023-01-01 00:16:41	2023-01-01 00:21:46	2.0	0.006053	1.0	N	161	237	1	...	Janu.
2	2	2023-01-01 00:14:03	2023-01-01 00:24:36	3.0	0.007029	1.0	N	237	141	2	...	Janu.
3	2	2023-01-01 00:24:30	2023-01-01 00:29:55	1.0	0.002636	1.0	N	143	142	2	...	Janu.
4	1	2023-01-01 00:42:56	2023-01-01 01:16:33	2.0	0.034658	1.0	N	246	37	1	...	Janu.

5 rows × 31 columns

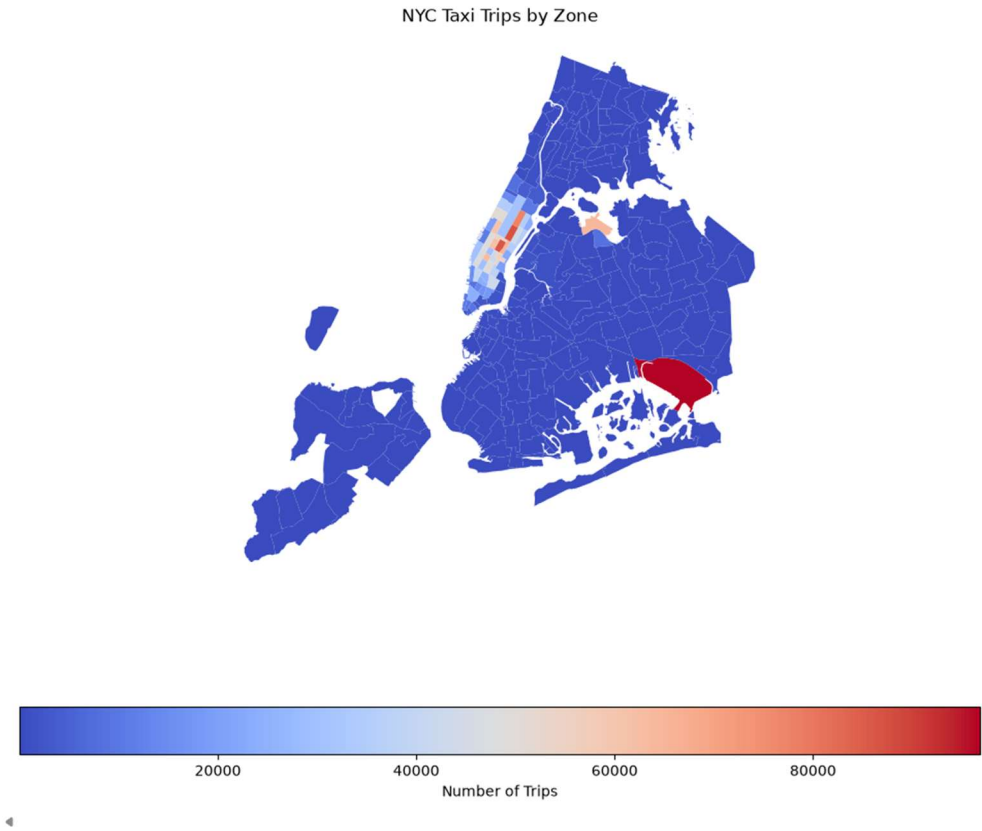
3.1.11. Find the number of trips for each zone/location ID

	PULocationID	num_trips
0	1	214
1	2	2
2	3	40
3	4	1861
4	5	13

3.1.12. Add the number of trips for each zone to the zones dataframe

	OBJECTID	Shape_Leng	Shape_Area	zone	LocationID	borough	geometry	PULocationID	num_trips
0	1	0.116357	0.000782	Newark Airport	1	EWB	POLYGON ((933100.918 192536.086, 933091.011 19...	1.0	214.0
1	2	0.433470	0.004866	Jamaica Bay	2	Queens	MULTIPOLYGON (((1033269.244 172126.008, 103343...	2.0	2.0
2	3	0.084341	0.000314	Allerton/Pelham Gardens	3	Bronx	POLYGON ((1026308.77 256767.698, 1026495.593 2...	3.0	40.0
3	4	0.043567	0.000112	Alphabet City	4	Manhattan	POLYGON ((992073.467 203714.076, 992068.667 20...	4.0	1861.0
4	5	0.092146	0.000498	Arden Heights	5	Staten Island	POLYGON ((935843.31 144283.336, 936046.565 144...	5.0	13.0

3.1.13. Plot a map of the zones showing number of trips



3.1.14. Conclude with results

Conclusion

- The analysis confirmed that trip distance is one of the main factors influencing fare amounts, with longer trips generally resulting in higher fares.
- Demand patterns vary by time and location, with the highest activity observed during rush hours, weekends, and in busy zones such as airports, transit hubs, and Midtown areas.
- Most taxi rides are taken by one or two passengers, and credit cards remain the most commonly used payment method.
- Data cleaning and preprocessing helped remove inconsistencies and outliers, resulting in a reliable dataset for uncovering meaningful trends and business insights.

3.2. Detailed EDA: Insights and Strategies

3.2.1. Identify slow routes by comparing average speeds on different routes

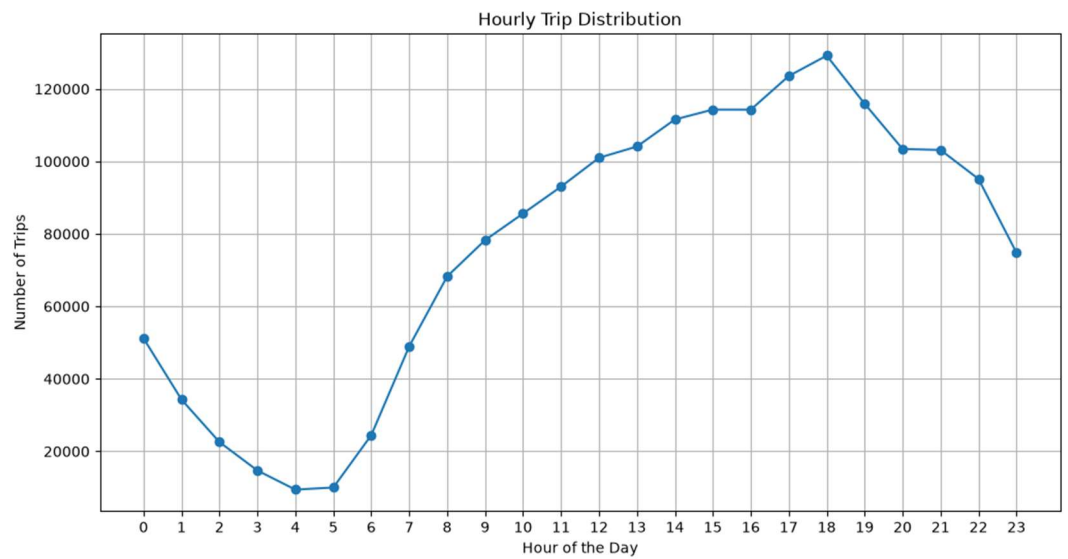
	PULocationID	DOLocationID	pickup_hour	avg_speed_mph
102294	232	65	13	0.000026
114929	243	264	17	0.000038
61252	142	142	5	0.000116
120428	258	258	1	0.000128
33393	100	7	8	0.000193
6451	40	65	21	0.000229
39490	113	235	22	0.000235
89226	194	194	16	0.000239
95261	226	145	18	0.000253
9705	45	45	10	0.000290



3.2.2. Calculate the hourly number of trips and identify the busy hours

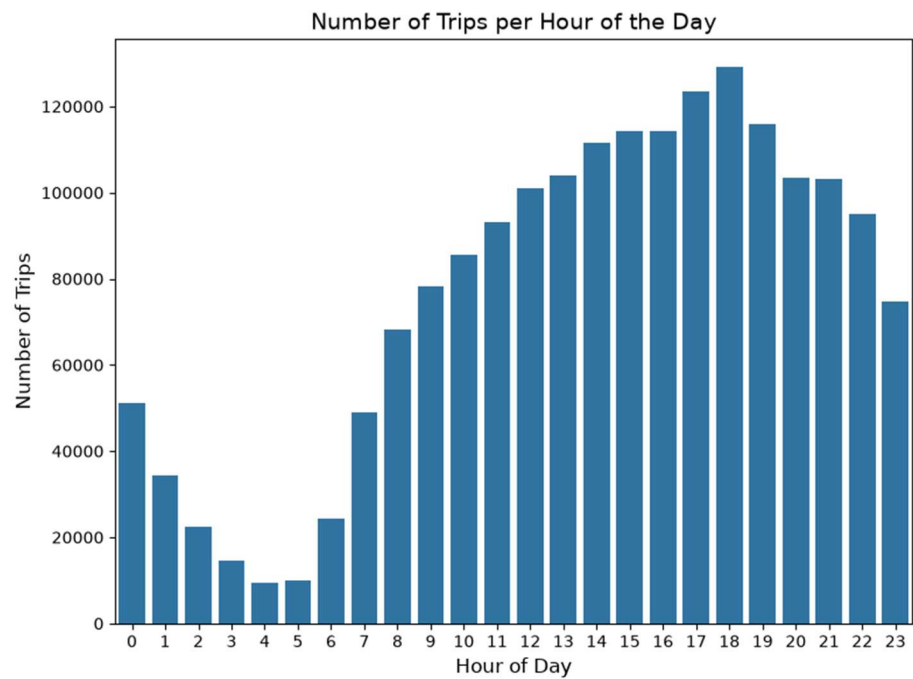
Busiest hour: 18

Number of trips during busiest hour: 129190



4

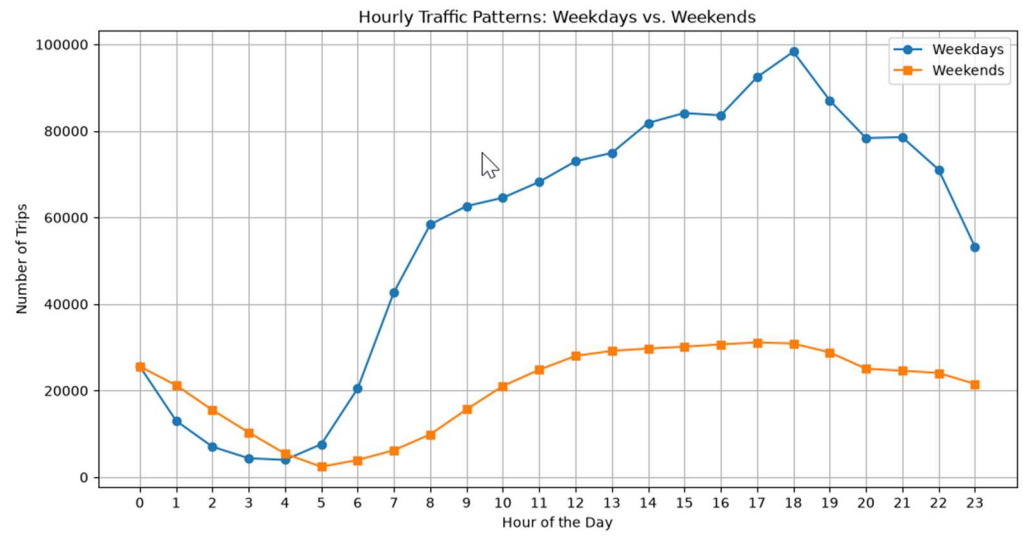
Busiest hour: 18
Number of trips during busiest hour: 129190



3.2.3. Scale up the number of trips from above to find the actual number of trips

Pickup Hour	Number of Pickups
18	129190
17	123563
19	115920
15	114301
16	114289

3.2.4. Compare hourly traffic on weekdays and weekends



3.2.5. Identify the top 10 zones with high hourly pickups and drops

Top 10 Pickup Zones:

	LocationID	Pickup_Trips	zone
0	132	96827	JFK Airport
1	237	86905	Upper East Side South
2	161	85948	Midtown Center
3	236	77517	Upper East Side North
4	162	65634	Midtown East
5	138	64177	LaGuardia Airport
6	186	63471	Penn Station/Madison Sq West
7	230	61315	Times Sq/Theatre District
8	142	60887	Lincoln Square East
9	170	54493	Murray Hill

Top 10 Dropoff Zones:

	LocationID	Dropoff_Trips	zone
0	236	81269	Upper East Side North
1	237	77558	Upper East Side South
2	161	71647	Midtown Center
3	230	56398	Times Sq/Theatre District
4	170	54314	Murray Hill
5	162	52248	Midtown East
6	142	51494	Lincoln Square East
7	239	51260	Upper West Side South
8	141	48449	Lenox Hill West
9	68	46352	East Chelsea

3.2.6. Find the ratio of pickups and dropoffs in each zone

```
top_pickup_dropoff_ratio
```

```
✓ 0.0s
```

```
zone
East Elmhurst      8.320717
JFK Airport        4.617626
LaGuardia Airport  2.884489
Penn Station/Madison Sq West 1.582187
Central Park       1.374760
Greenwich Village South 1.374743
West Village       1.326222
Midtown East       1.256201
Midtown Center     1.199604
Garment District   1.191880
Name: pickup_dropoff_ratio, dtype: float64
```

```
bottom_pickup_dropoff_ratio
```

```
✓ 0.0s
```

```
zone
Freshkills Park    0.000000
Oakwood            0.000000
Broad Channel      0.000000
West Brighton      0.000000
Breezy Point/Fort Tilden/Riis Beach 0.025641
Stapleton          0.029412
Windsor Terrace    0.038259
Newark Airport     0.040233
Grymes Hill/Clifton 0.043478
Ridgewood          0.052525
Name: pickup_dropoff_ratio, dtype: float64
```

3.2.7. Identify the top zones with high traffic during night hours

```
top_night_pickups
```

pickup_zone	
East Village	15339
JFK Airport	13399
West Village	12352
Clinton East	9797
Lower East Side	9535
Greenwich Village South	8720
Times Sq/Theatre District	7776
Penn Station/Madison Sq West	6233
Midtown South	5962
LaGuardia Airport	5947
Name: PULocationID, dtype: int64	

3.2.8. Find the revenue share for nighttime and daytime hours

Nighttime Revenue Share: 12.06%

Daytime Revenue Share: 87.94%

- 3.2.9. For the different passenger counts, find the average fare per mile per passenger

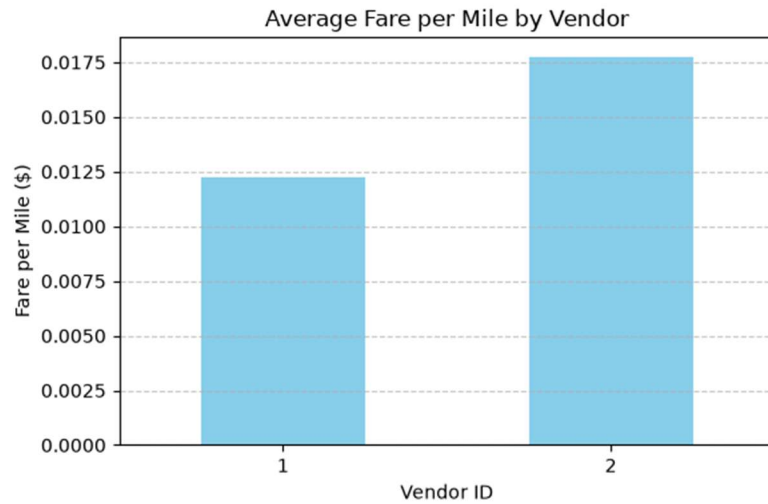
	passenger_count	Average_Fare_Per_Mile_Per_Passenger
0	1.000000	0.024175
1	2.000000	0.013309
2	3.000000	0.008308
3	4.000000	0.008498
4	5.000000	0.003936
5	6.000000	0.003173

- 3.2.10. Find the average fare per mile by hours of the day and by days of the week

	day_of_week	fare_by_day
0	Friday	0.02
1	Monday	0.02
2	Saturday	0.02
3	Sunday	0.03
4	Thursday	0.02
5	Tuesday	0.03
6	Wednesday	0.02

	hour_of_day	fare_by_hour
0	0	0.02
1	1	0.02
2	2	0.02
3	3	0.02
4	4	0.03
5	5	0.03
6	6	0.02
7	7	0.02
8	8	0.02
9	9	0.02
10	10	0.03
11	11	0.02
12	12	0.02
13	13	0.02
14	14	0.02
15	15	0.03
16	16	0.03
17	17	0.03
18	18	0.03
19	19	0.03
20	20	0.02
21	21	0.02
22	22	0.02
23	23	0.02

3.2.11. Analyse the average fare per mile for the different vendors



Vendor 2 earns approximately 45% more fare per mile than Vendor 1, suggesting an opportunity to further investigate the underlying causes of this pricing difference.

3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion



- For trips up to 2 miles, Vendor 2 charges significantly more per mile ($\approx \$17.9$) than Vendor 1 ($\approx \10.6), indicating a stronger revenue contribution from short-distance trips.
- For 2 to 5 miles and more than 5 miles, the fare per mile drops considerably for both vendors, suggesting that longer trips benefit from economies of scale and lower cost per mile.
- Vendor 2 consistently has a slightly higher fare per mile across all distance tiers, indicating a more profitable pricing structure compared to Vendor 1.
- Short-distance trips generate the highest fare per mile for both vendors, making them the most lucrative trip category from a revenue perspective.

3.2.13. Analyse the tip percentages

```
Average Tip Percentage by Distance:
distance_category
Up to 2 miles      12.144307
2 to 5 miles      12.289765
More than 5 miles  11.307092
Name: tip_percentage, dtype: float64
```

```
Average Tip Percentage by Passenger Count:
passenger_category
1 passenger      12.174914
2-3 passengers   11.700981
4+ passengers    11.348378
Name: tip_percentage, dtype: float64
```

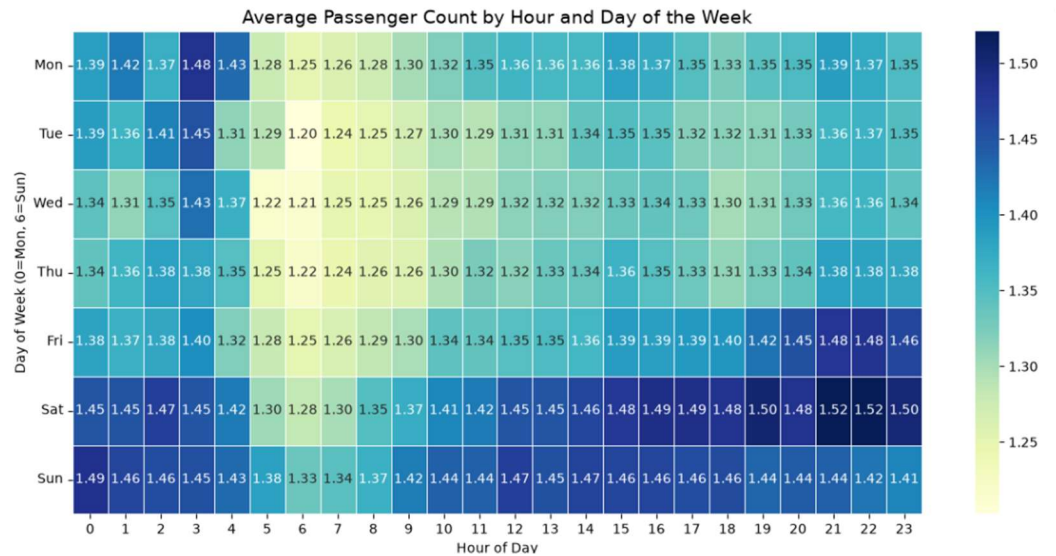
```
Average Tip Percentage by Time of Pickup:
time_category
Midnight to 6 AM    11.659648
6 AM to Noon        11.902430
Noon to 6 PM        11.860486
6 PM to Midnight    12.405365
Name: tip_percentage, dtype: float64
```

```
Most Common Low Tip Scenarios:
distance_category  passenger_category  time_category  count
Up to 2 miles      1 passenger        Noon to 6 PM    97300
...
Up to 2 miles      2-3 passengers    6 PM to Midnight  25233
Up to 2 miles      2-3 passengers    Noon to 6 PM     24707
Up to 2 miles      2-3 passengers    6 PM to Midnight  22340
```

Analysis: Tip Percentage Patterns

- Medium-distance trips (2 to 5 miles) have the highest average tip percentage (12.29%), while trips longer than 5 miles receive slightly lower tips (11.31%).
- Solo travelers tend to tip more (12.17%) than groups, with tip percentages decreasing as passenger count increases.
- Evening trips (6 PM to Midnight) have the highest average tip percentage (12.41%), suggesting that passengers are generally more generous during leisure and entertainment hours.
- The most common low-tip scenarios occur for short trips (up to 2 miles), particularly with one passenger during the afternoon (Noon to 6 PM), indicating that short-distance rides are more likely to receive lower tips despite their high frequency.

3.2.14. Analyse the trends in passenger count



Key Observations

Weekend trips have the highest average passenger counts, with Saturday evenings (9 PM to 10 PM) reaching the peak value of approximately 1.52 passengers per trip.

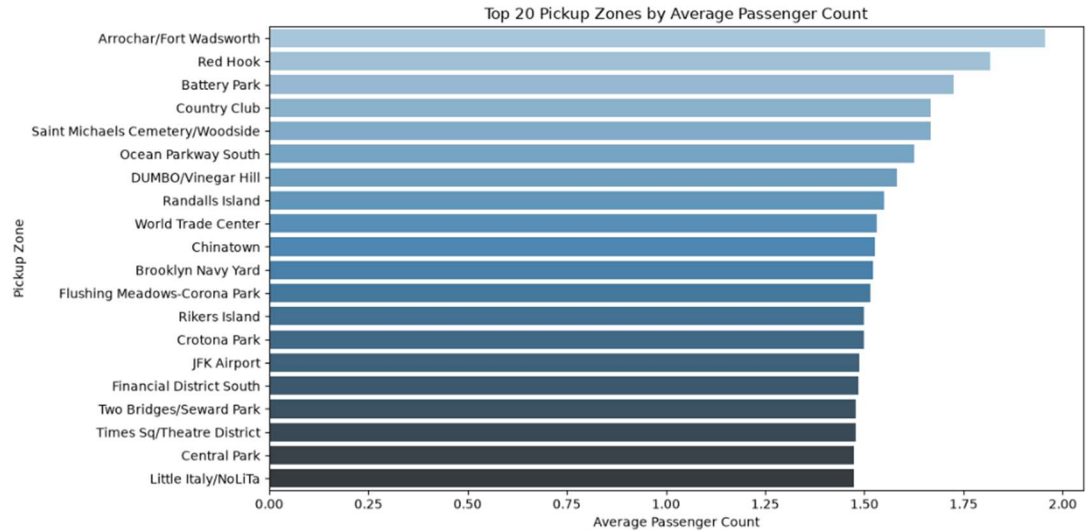
Weekday morning hours (5 AM to 9 AM) have the lowest average passenger counts, ranging from 1.20 to 1.28, indicating that most trips are taken by individual commuters.

Late-night and early-morning hours (12 AM to 4 AM) consistently show higher passenger counts across most days, suggesting increased group travel during nightlife and entertainment hours.

Friday evenings mark the start of weekend travel behavior, with passenger counts increasing steadily after 6 PM.

Overall, passenger counts tend to increase during afternoons and evenings and decrease during weekday morning hours.

3.2.15. Analyse the variation of passenger counts across zones



Key Observation:

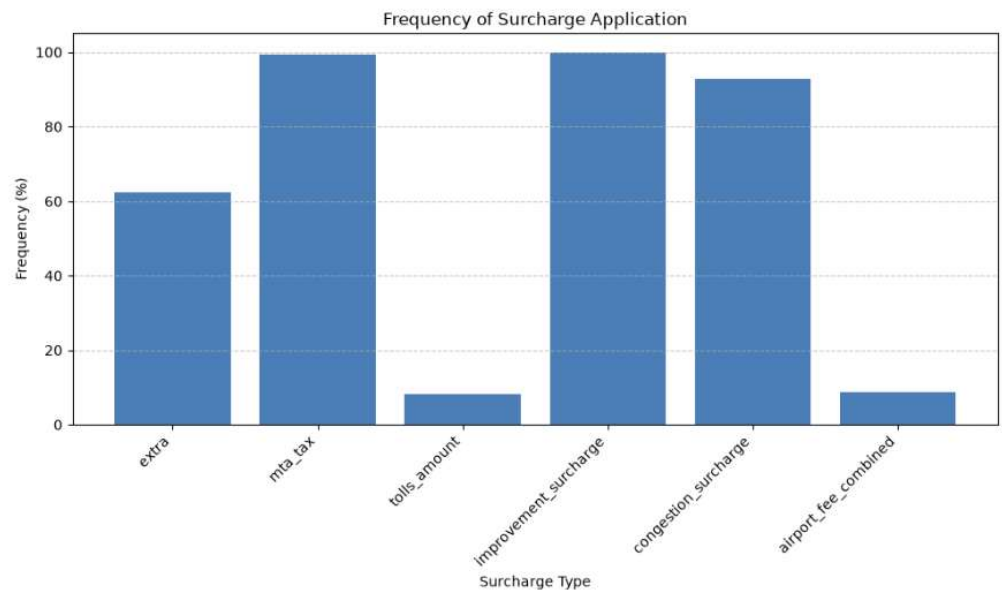
Tourist and leisure destinations tend to generate more multi-passenger rides than standard commuting areas.

Airport-related pickups are associated with larger travel groups.

3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.

Surcharge Analysis:

	Frequency (%)	Average Amount (\$)
extra	62.31	1.65
mta_tax	99.36	0.50
tolls_amount	8.10	0.00
improvement_surcharge	99.99	1.00
congestion_surcharge	92.92	2.32
airport_fee_combined	8.78	0.14



Observation:

Congestion surcharge is the most impactful surcharge from a revenue perspective because it is both highly frequent and relatively high in value.

Extra charges represent another source of revenue, affecting more than half of all trips.

Fixed charges such as the MTA tax and improvement surcharge provide a consistent revenue stream across nearly all trips.

4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

Demand Patterns

- Taxi demand is highest during rush hours, weekends, and certain times of the year.
- Nightlife and entertainment districts experience strong demand during evening and late-night hours.
- Airports, transportation hubs, tourist attractions, and business districts consistently generate a large number of trips.

Fare and Revenue Trends

- Trip fares generally increase with distance and travel time.
-

Location-Based Trends

- Some pickup and drop-off zones experience significantly higher demand than others.
- Popular destinations and entertainment hotspots attract a high volume of trips, especially during evenings and weekends.

Pricing and Surcharges

- Surcharges such as congestion fees, improvement surcharges, and MTA taxes are applied to most trips and contribute significantly to revenue.
- Fare structures vary depending on trip type, distance, and location.
- Distance-based pricing remains an important factor in overall revenue generation.

Recommendations for Improvement

Improve Demand Coverage

- Prioritize taxi availability in high-demand zones during peak hours.
- Increase service coverage in nightlife districts during evening and late-night periods.
- Encourage ride-sharing options in areas with frequent group travel.

Optimize Fleet Deployment

- Position more taxis in locations with consistently high demand, such as airports, business districts, and tourist attractions.
- Use demand forecasts to proactively allocate vehicles throughout the city.
- Encourage drivers to relocate toward underserved areas through targeted incentives.

- Consider dynamic pricing during periods of exceptionally high demand to balance supply and demand.

Conclusion

- Taxi demand is highest during rush hours, weekends, airport trips, and in major business and entertainment zones, making these areas key focus points for fleet deployment.
- Trip distance, travel time, passenger count, and pickup location have a significant impact on fares, tips, and overall revenue generation.
- Congestion surcharges, MTA taxes, and improvement surcharges are applied to most trips and contribute consistently to taxi revenue.
- Optimizing taxi availability in high-demand areas, promoting ride-sharing, and using demand-based planning can improve both operational efficiency and customer satisfaction.

4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.

Time-Based Planning

- Increase cab availability during morning and evening rush hours when passenger demand is at its highest.
- Deploy additional taxis near nightlife and entertainment districts during late evenings and weekends.
- Reduce idle time by reallocating cabs from low-demand areas during midday periods.
- Use historical monthly trends to prepare for seasonal increases in demand.

Day-Based Planning

- Position more cabs in commercial and business districts on weekdays to support commuter traffic.
- Increase coverage around tourist attractions, shopping areas, and entertainment zones during weekends.
- Allocate additional vehicles on Fridays and Saturdays, when demand tends to remain high throughout the evening.

Zone-Based Planning

- Maintain higher taxi availability at airports, transit hubs, and other consistently busy pickup locations.
- Monitor zones with high pickup activity but limited taxi availability and prioritize fleet deployment in those areas.

- Reposition taxis from low-demand zones to nearby high-demand locations to improve utilization and reduce passenger wait times.

Data-Driven Fleet Management

- Continuously monitor trip patterns and demand fluctuations across zones.
- Use demand forecasts to determine where taxis should be positioned throughout the day.
- Regularly review operational data and adjust deployment strategies to match changing travel patterns and customer needs.

Conclusion

- Focus cab deployment on high-demand areas such as airports, business districts, tourist attractions, and entertainment hubs.
- Align fleet availability with demand patterns across different times of the day, days of the week, and seasons.
- Proactively reposition taxis to reduce idle time and improve service coverage in busy zones.
- Use historical and real-time trip data to make informed dispatching decisions and improve overall operational efficiency.

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

- **Surcharge Optimization:** Review surcharge usage regularly and apply additional charges only when demand justifies them. Clear communication about these charges helps build customer trust and avoid surprises.
- **Stay Competitive:** Keep track of market pricing and competitor offerings. Adjust fares when needed while emphasizing service quality, convenience, and reliability to differentiate your services.
- **Continuous Improvement:** Regularly analyse trip and revenue data, test different pricing approaches, and refine strategies based on customer behaviour and operational performance.

By making informed pricing decisions and continuously monitoring market trends, taxi companies can improve revenue, remain competitive, and provide a better overall experience for passengers.